# Oxygen Therapy in Office Practice

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## Introduction

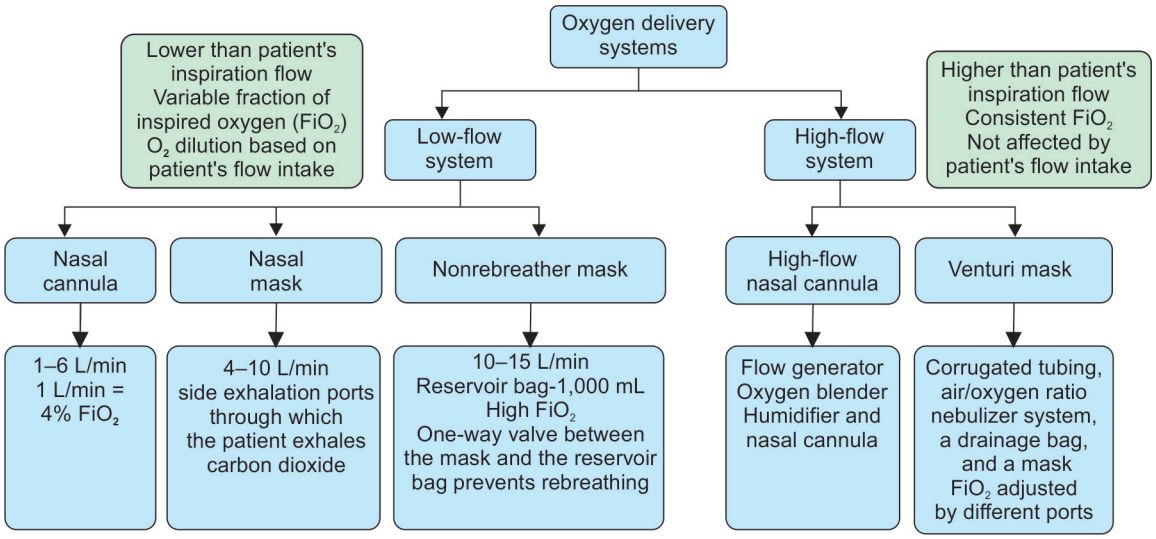
- ☑ Oxygen therapy is the administration of oxygen at a concentration and pressure greater than that found in the environmental atmosphere to treat hypoxemia.
- ☑ The cost, benefits, and risks of oxygen therapy should be considered as any other drug taking due care to avoid excessive as well as inadequate dosing.

## Indications of Oxygen Initiation

Any oxygen saturation ($SpO_2$) < 90% needs oxygen therapy. Common situations needing oxygen therapy are summarized here.

| *System* | *Mild to moderate* | *Severe* |
|---|---|---|
| ☑ Central nervous system | ☑ Confusion, agitation, combativeness | ☑ Lethargy, obtunded mental status |
| ☑ Cardiac | ☑ Tachycardia, ectopy, hypertension | ☑ Bradycardia, hypotension |
| ☑ Respiratory | ☑ Dyspnea, tachypnea, shallow respirations, labored breathing | ☑ Increasing dyspnea and tachypnea, possible bradypnea or agonal respirations |
| ☑ Skin | ☑ Cool, clammy | ☑ Cyanosis |
| ☑ Arterial blood gas | ☑ $PaO_2$: 60–80 mm Hg | ☑ $PaO_2$: <60 mm Hg |

## Classification of Oxygen Delivery Systems




## Common Types of Oxygen Devices and Its Usage

| *Device* | *Flow and $FiO_2$* | *Indications* | *Advantages* | *Disadvantages* |
|---|---|---|---|---|
| Free-flow oxygen | 3–4 L/min $FiO_2$: 30–40% | Minimal oxygen requirement in irritable child | Least invasive of providing oxygen | Unpredictable oxygen delivery |
| Nasal cannula | 1–4 L/min $FiO_2$: 24–40% | Minimal oxygen requirement Recovery phase from acute episodes | ☑ Low cost ☑ Well tolerated ☑ No rebreathing | ☑ Uncontrolled oxygen delivery ☑ Dries the nose |
| Facemask | 5–10 L/min $FiO_2$: 40–60% | Used for children with moderate oxygen requirement | Useful in children who are mouth breathers, those with nasal irritation or epistaxis | Minimum delivery 5 L/min to avoid rebreathing of expired $CO_2$ |

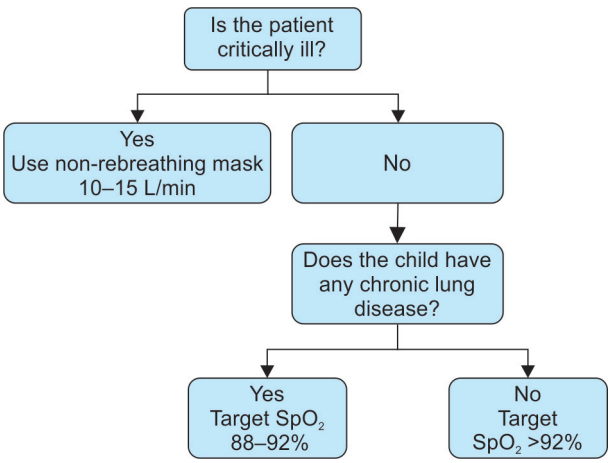
*Contd…*

## Common Types of Oxygen Devices and Its Usage

*Contd…*

| Device | Flow and $FiO_2$ | Indications | Advantages | Disadvantages |
|---|---|---|---|---|
| Non-rebreather mask | 10–15 L/min<br>$FiO_2$: 60–100% | ☑ Used for patients who are hypoxic and in respiratory distress that require high $FiO_2$<br>☑ Recommended as the device of choice to deliver 100% oxygen in the advanced life support guidelines | Can deliver close to 100% oxygen | Requires tight seal on the face which may cause discomfort |
| Oxygen hood | 3–7 L/min (2–3 mL/kg/min)<br>$FiO_2$: 21–100% | Newborn/young infants | ☑ Well tolerated<br>☑ Can deliver high concentrations of oxygen<br>☑ Easy access for other care | ☑ Inadequate flow rate can result in rebreathing<br>☑ Hard to maintain uniform oxygenation during nursing maneuvers<br>☑ High flow needed leads to wastage of $O_2$ |
| High-flow nasal cannula | 2–60 L/min<br>$FiO_2$: 21–100% | Useful in conditions with mild-to-moderate respiratory distress of lower airway origin like bronchiolitis | ☑ Comfortable and well tolerated<br>☑ Can accurately titrate concentrations of oxygen<br>☑ Can feed through mouth | ☑ Cannula is expensive<br>☑ Needs high-flow delivery device<br>☑ Needs humidification device<br>☑ Used in in-hospital setting |

## Initiation of Oxygen Therapy



## Types of Oxygen Cylinders

| *Cylinder size* | *Capacity (L)* | *Pressure (psi)* |
|---|---|---|
| E (Anesthesia work stations) | 660 | 1,900 |
| F (Ward or transport) | 1,360 | 1,900 |
| H (Cylinder manifold) | 6,900 | 2,200 |

## General Principles of Managing Oxygen Cylinders

- ☑ The cylinders should be stored in a cool, dry, and well-ventilated area which has been constructed with fire-resistant material.
- ☑ Empty and full cylinders should be stored separately and never in direct sunlight.
- ☑ There should be a minimum of 2-day supply in each cylinder and a supply of at least 3 days of spare cylinders kept reserve in the manifold room.

☑ Calculating duration $O_2$ cylinder will last:

☑ $$\frac{Pr \times V ri}{Pi \times Q \times 60}$$

- $Tr$ = Time remaining in hours
- $Pr$ = Remaining pressure
- $V$ = Initial volume of O available in the full cylinder *i*2
- $Pi$ = Initial pressure in the full cylinder (approximately 2000 psi or 140 × 100 kPa)
- $Q$ = Flow rate of $O_2$ required.

- ☑ Always make sure the valves are closed and cylinder caps in place.
- ☑ Never apply excessive force to cylinder valves; so do not lift or carry a gas cylinder using their valves, shrouds, or caps.
- ☑ Always use a trolley or lifting device (like a forklift) for moving cylinders.
- ☑ Because a cylinder must always be protected from being knocked over, falling, or impact damage to the valve; make sure the trolley or lifting device has secure restraints and safety straps.
- ☑ Set up the cylinder for patient use before placing it close to the patient. This includes connecting the appropriate equipment, opening the valve (slowly), and selecting the flow.
- ☑ Avoid placing the cylinder on the bed next to the patient if at all possible; use extra care when there is no option but to place the cylinder on the bed ensuring the cylinder has been turned on first and the flow selected.

## Monitoring of Oxygen Therapy

☑ Basic principles:

- Monitor minimum 4 hourly
- Titrate and wean off oxygen
- Always give the minimum oxygen required

☑ What to monitor?

- Monitor cardiopulmonary status, including vital signs, $SpO_2$, and indicators of oxygenation and ventilation (clinical monitoring of respiratory efforts, respiratory rate, and level of consciousness).
- Monitor the child's respiratory status and need for increased or decreased oxygen therapy.
- Monitor the child for signs of hypercarbia (increasing agitation, rapid and deep respiration, dyspnea, and progressive lethargy).
- Assess the skin frequently for breakdown and for signs of dry mucous membranes.
- Use a recording sheet like the one given here is recommended:

| | Target saturation | ☐ 88–92 | ☐ 92–94 | ☐ 94–98 | | |
|---|---|---|---|---|---|---|
| Device/ interface | Flow rate | Respiratory rate (RR) | $SpO_2$ | Heart rate (HR) | Start time | End time |
| | | | | | | |
| | | | | | | |
| | | | | | | |

## Titration of Oxygen Therapy

Venturi 24% 2–4 L/min (blue) or nasal cannula 1 L/min

↕

Venturi 28% 4–6 L/min (white) or nasal cannula 2 L/min

↕

Venturi 35% 8–10 L/min (yellow) or nasal cannula 4 L/min

↕

Venturi 40% 10–12 L/min (red) or simple face mask 5–6 L/min

↕

Venturi 60% 12–15 L/min (green) or simple face mask 7–10 L/min

↕

Non-rebreathing mask @15 L/min

*Note:* Escalation of $O_2$ support requires senior medical input immediately

**Signs of deterioration Increased**
- RR
- $FiO_2$ to reach target $SpO_2$
- $CO_2$ retention signs
- Drowsiness

Decreased
- $SpO_2$
- Respiratory efforts

**Key tips:**
- Increased $O_2$ requirement should be reviewed by senior medical person within 30 minutes
- Escalate to respiratory support if $O_2$ requirement crosses 60%

## End-Points Oxygen Therapy

- ☑ Previously well children recovering from acute illness—$SpO_2$ > 92%
- ☑ Child recovering from acute illness with background of chronic lung disease/cystic fibrosis/pulmonary hypertension—$SpO_2$ >88% in room air
- ☑ Children with cyanotic congenital heart disease—their previous baseline saturation.

## Dangers of Oxygen Therapy

- ☑ Absorption atelectasis
- ☑ Free radical injury.

## Further Reading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- ☑ Sarkar PK, Rahman MS. Oxygen therapy in children: an update. DS (Child) H J. 2018;34:48-54.
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Indian Academy of Pediatrics (IAP)

# STANDARD TREATMENT GUIDELINES 2022

# Staphylococcal Scalded Skin Syndrome


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